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Statement of Purpose
of Gerard Lawler

My primary interests lie in the physics of surfaces in novel environments especially from the perspective of electron emission. To this end my, my graduate school education was primarily concerned with the impacts of low temperatures and high electromagnetic fields on electron emission in the context of linear accelerators. My work was primarily funded by the National Science Foundation Center for Bright Beams (NSF CBB), one of the main goals of which is to improve beam brightness via improving electron source performance at the cathode. Improving cathode performance by various mechanisms is then indeed the area of physics which I intend to continue studying for my future career.

Throughout the course of working on my masters and later PhD thesis, my work involved two main projects: the development and testing of nanofabricated metallic field emission cathodes and the design and commissioning of a new high gradient cryogenic RF photogun to act as a cathode test bed beamline at UCLA.

Nanofabricated cathodes: I worked on the development, nanofabrication and, testing of patterned nanoblade cathodes in order to produce high brightness field electron beams. The approach taken was from both an experimental and theoretical direction. In general, the idea is to produce nanopatterned field emitters arrays via geometric laser field enhancement from atomically sharp structures etched onto silicon wafers. The atomically sharp structures are then coated with nm thick layers of metal longer than the laser field skin depth. Thus far we have produced gold and tungsten coated samples for study. The nm scale structures reduce the effective laser spot size on a cathode reducing the overall mean transverse energy (MTE). The unique feature here is that the structures themselves are 1D nanoblades rather than tip emitters are far more robust with respect to laser fluence (Mann, Lawler, and Rosenzweig (2019)).

In order to measure these cathodes I also developed a cost effective hemispherical deflection analyzer. Some of the results are reported in the following publication (G. Lawler et al. (2019)). During the course of the work we also experimented with modification of the nanostructures with focused ion beams (FIB).

Cryogenic cathode test bed: The bulk of my PhD thesis involved the design and commissioning of a new high gradient cryogenic RF photogun specifically for testing of cathodes in extreme environments. There are competing advantages and disadvantages for cathodes operated at cryogenic temperatures (≤ 120 K) and high cathode field gradients (≥ 100 MV/m). Cryogenic temperatures can positively reduce the MTE but also negatively reduce the quantum efficiency (QE). High cathode launch fields can improve beam dynamics but also damage sensitive cathodes. My work involved considering these competing effects in the context of metallic photocathodes (J. Rosenzweig et al. (2018)). For the sake of improving the electron beam brightness it then became useful to develop a test facility for this novel environment.

To this end, I developed and commissioned a unique CrYogenic Brightness-Optimized Radiofrequency Gun (CYBORG) to fulfill the needs of testing metallic and other advanced cathodes such as semiconductors in extreme conditions. The system was designed for compatibility with a load-lock for cathode transfer and has reached 85K with peak launch field in the range $95-105$ MV/m as intended (G. E. Lawler et al. (2024)). In addition, the design serves as a template for a new XFEL photoinjector (J. B. Rosenzweig et al. (2020)). Through my participation in the NSF CBB I also collaborated with superconducting RF groups on cryogenics development.

Conclusion: These two projects highlight my competence in the field of cathode design. Specifically, I am aware of the necessary features in producing cathodes for high brightness applications such as XFELs and well versed in the capabilities of the types of photogun environments in which these cathode live. When combined with my future career goals of continuing advancements in the understanding of electron emission in novel environments, this makes me an excellent candidate for the position in photocathode development at DESY for the European XFEL.

References

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